

EX. NO: 6

RAIL FENCE CIPHER

Aim:

To write a C program to implement the Rail Fence Cipher, a transposition technique that rearranges plaintext characters in a zig-zag pattern across a defined number of rails.

Algorithm:

1. Start the program.
2. Read the plaintext and the number of rails from the user.
3. Create a two-dimensional matrix with rows equal to the number of rails and columns equal to the length of the plaintext.
4. Traverse the plaintext and place each character into the matrix, moving down and up across the rails in a zig-zag manner.
5. Read the matrix row by row, skipping empty cells, to obtain the encrypted text.
6. Display the resulting encrypted text.
7. Stop the program.

Program:

```
#include <stdio.h>
#include <string.h>

int main() {
    char text[100];
    int key;

    printf("Enter Plain Text: ");
    scanf("%s", text);
    printf("Enter Number of Rails: ");
    scanf("%d", &key);

    int len = strlen(text);
    char rail[key][len];

    for (int i = 0; i < key; i++)
        for (int j = 0; j < len; j++)
            rail[i][j] = '\n';

    int row = 0;
    int dir = 1;
    for (int i = 0; i < len; i++) {
        rail[row][i] = text[i];
        if (row == 0)
            dir = 1;
        else if (row == key - 1)
            dir = -1;
        row += dir;
    }

    printf("Encrypted Text: ");
    for (int i = 0; i < key; i++)
        for (int j = 0; j < len; j++)
            if (rail[i][j] != '\n')
                printf("%c", rail[i][j]);
    printf("\n");
}
```

```

        dir = -1;
        row += dir;
    }

    printf("Encrypted Text: ");
    for (int i = 0; i < key; i++)
        for (int j = 0; j < len; j++)
            if (rail[i][j] != '\n')
                printf("%c", rail[i][j]);
    printf("\n");
    return 0;
}

```

Output:

```

1 #include<stdio.h>
2 #include<string.h>
3
4 int main()
5 {
6     char text[100];
7     int key;
8
9     printf("Enter Plain Text: ");
10    scanf("%s",text);
11
12    printf("Enter Number of Rails: ");
13    scanf("%d",&key);
14
15    int len=strlen(text);
16
17    char rail[key][len];
18
19    for(int i=0;i<key;i++)
20        for(int j=0;j<len;j++)
21            rail[i][j]='\n';

```

```

Enter Plain Text: THANVEER
Enter Number of Rails: 2
Encrypted Text: TAVEHNER

=== Code Execution Successful ===

```

Result:

Thus a C program to implement the Rail Fence Cipher transposition technique was written, executed and the output was verified successfully.